

Lab # 03

Implementation of Relational Database Model using Query Language

Objective:

Student will learn about to do the following:

- Learn the data definition language commands to create the table.

Software Tools:

- MySQL Workbench

Theory:

MySQL Workbench is a visual database design and management tool developed by Oracle. It is a powerful and popular integrated development environment (IDE) for working with MySQL and other compatible relational database management systems (RDBMS).

MySQL Workbench offers a wide range of features and functionalities, making it a versatile tool for database professionals, developers, and administrators. Some of its key features include:

Database Design: You can use MySQL Workbench to create, edit, and visualize database schemas using its Entity-Relationship Diagram (ERD) modeling tool. This makes it easier to design and understand the structure of your databases.

SQL Development: MySQL Workbench provides a SQL editor with features like syntax highlighting, code completion, and query execution. It allows you to write and test SQL queries and scripts interactively.

Database Administration: It offers tools for managing and administering databases, including user and privilege management, server monitoring, and backup and restore operations.

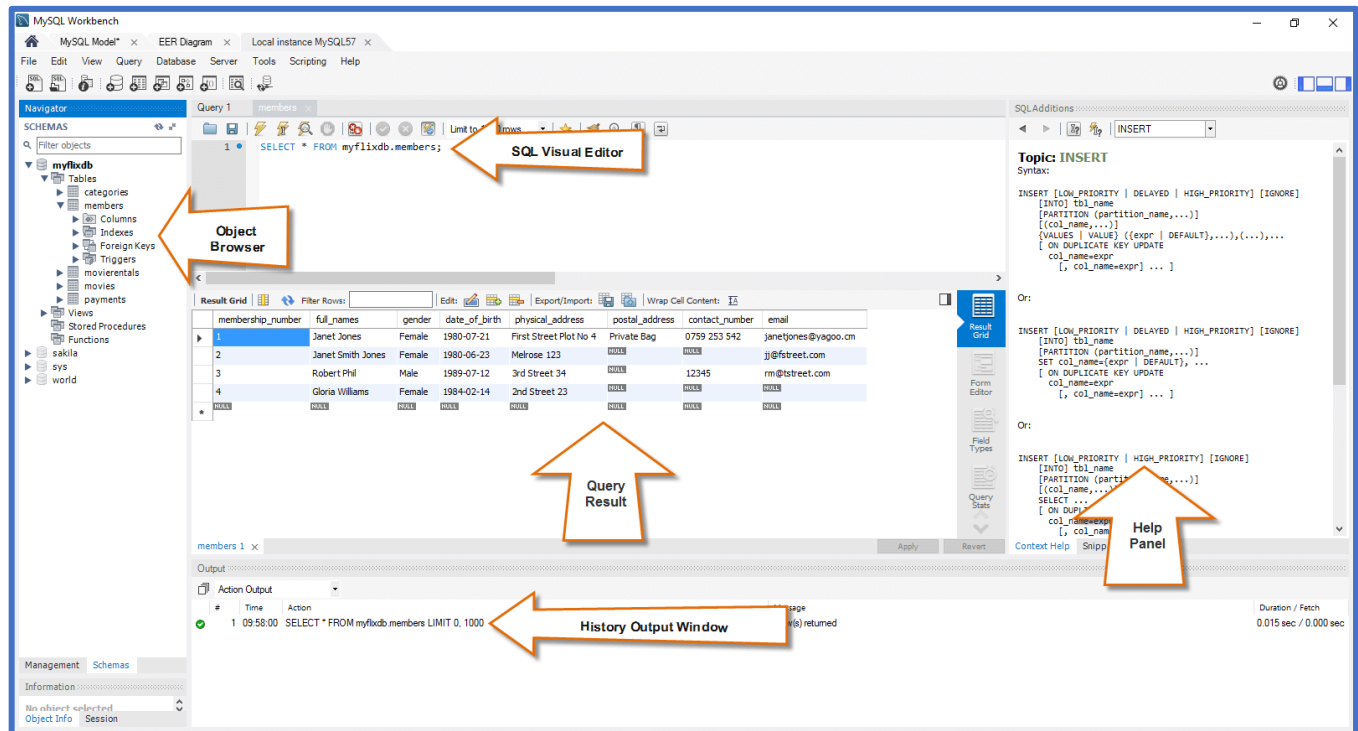
Data Modeling: You can forward- and reverse-engineer database designs to and from SQL code, making it easy to transition between a visual model and the actual database schema.

Migration and Synchronization: MySQL Workbench can assist in migrating data from one database to another, and it supports schema synchronization to keep databases in sync with their models.

Performance Analysis: The tool includes performance monitoring and analysis features to help identify and optimize slow-performing queries and database operations.

Scripting and Automation: You can automate various database-related tasks by creating and executing scripts using Python or SQL.

Reporting: It offers tools for generating detailed reports and documentation about your database structures.



SQL is a structured query language. SQL contains different data types those are

Data type	Description
CHARACTER(n)	Character string. Fixed-length n
VARCHAR(n)	Character string. Variable length. Maximum length n
BOOLEAN	Stores TRUE or FALSE values
VARBINARY(n)	Binary string. Variable length. Maximum length n
INTEGER(p)	Integer numerical (no decimal). Precision p
SMALLINT	Integer numerical (no decimal). Precision 5

INTEGER	Integer numerical (no decimal). Precision 10
BIGINT	Integer numerical (no decimal). Precision 19
DECIMAL(p, s)	Exact numerical, precision p, scale s. Example: decimal(5,2) is a number that has 3 digits before the decimal and 2 digits after the decimal
NUMERIC(p, s)	Exact numerical, precision p, scale s. (Same as DECIMAL)
FLOAT(p)	Approximate numerical, mantissa precision p. A floating number in base 10 exponential notation. The size argument for this type consists of a single number specifying the minimum precision
REAL	Approximate numerical, mantissa precision 7
FLOAT	Approximate numerical, mantissa precision 16
BINARY(n)	Binary string. Fixed-length n

Different types of commands in SQL:

1. **DDL commands:** - To create a database objects
2. **DML commands:** - To manipulate data of a database objects
3. **DQL command:** - To retrieve the data from a database.
4. **DCL/DTL commands:** - To control the data of a database...

DDL Commands

DDL stands for **DATA DEFINITION LANGUAGE**. These are the statements that permit database tables to be created or deleted. It also defines indexes, specify links between tables and impose constraints between tables.

Examples Of DDL:

Commands	Working
CREATE DATABASE	It creates a new database.
ALTER DATABASE	It modifies the database.
CREATE TABLE	It creates a new table.
ALTER TABLE	It modifies the table
CREATE INDEX	Creates an index.
DROP INDEX	Deletes an index

Constraints

Constraints are used to limit the type of data that can go into a table. Constraints can be specified when a table is created (with the CREATE TABLE statement) or after the table is created (with the ALTER TABLE statement). We will focus on the following constraints:

o NOT NULL o UNIQUE o PRIMARY KEY o FOREIGN KEY o CHECK o DEFAULT

Note: We will use semicolons at the end of each SQL statement.

Queries and Commands for DDL

Create Database Command: To create a new database, you can use the **CREATE DATABASE** statement.

Syntax:

Create database <database name>;

Ex:

```
CREATE DATABASE WXYZ;    -- Creates a database named WXYZ;
```

Are you able to see the database created...???

If no, then refresh the services again...!!

Create Table Command: It defines each column of the table uniquely. Each column has a minimum of three attributes, a name, data type and size.

Syntax:

Create table <table name> (<col1> <datatype>(<size>),<col2> <datatype><size>));

Ex:

```
CREATE TABLE Employee (employ_id number(4) primary key, emp_name char(10));
```

Are you able to see the table created...???

Modifying the structure of tables: Modifying tables involves adding new columns.

Syntax:

Alter table <tablename> **add** <new col><datatype(size),<new col>datatype(size);

Ex: Alter table Employee **ADD** sal number(10);

Alter table Employee **ADD COLUMN** sal number(10);

Renaming the tables: Changing the name of a table.

Syntax:

Alter table <old_table>

Rename to <new_table>;

Ex:

```
ALTER TABLE emp1 RENAME TO emp2;
```

Destroying tables: It effectively deletes the entire table and all its contents, including all rows of data and the table structure.

Syntax:

Drop table <tablename>;

Ex:

```
DROP TABLE emp;
```

LAB TASK:

- A. Design a database system for a library inventory system. The library recently received a large collection of books and would like to keep better track of the titles. The library is still using an antiquated method for cataloging most of its books. Because of your computer skills, you have been asked to create a database containing the catalog number, title, author, and copyright date of each of the new books. Fill NULL in blank places.

Catalog #	Title	Description	Acquired Date	Location	ISBN	Condition
1	Speak	Paperback	8/12/2006	Building	340817623	Binding loose
3	Hoot	Paperback	8/17/2006	Building	375821813	
4	Artemis Fowl	Hardcover	8/17/2006	Bookmobile	141312122	Missing dust jacket
5	Wolfie	Hardcover	9/29/2008	Bookmobile	299389231	
6	Student Name	Paperback	9/29/2008	Bookmobile	55555555	

- B. Design the following database using SQL commands.

Special_Offer_ID	ETL_Source	Offer_Name	DiscountPct	Type	Category
1	ADVENTURE	No Discount	0.00	No Discount	No Discount
2	ADVENTURE	Volume Discount 11 to 14	0.02	Volume Discount	Reseller
3	ADVENTURE	Volume Discount 15 to 24	0.05	Volume Discount	Reseller
4	ADVENTURE	Volume Discount 25 to 40	0.10	Volume Discount	Reseller
5	ADVENTURE	Volume Discount 41 to 60	0.15	Volume Discount	Reseller
6	ADVENTURE	Volume Discount over 60	0.20	Volume Discount	Reseller
7	ADVENTURE	Mountain-100 Clearance Sale	0.35	Discontinued Product	Reseller
8	ADVENTURE	Sport Helmet Discount-2002	0.10	Seasonal Discount	Reseller
9	ADVENTURE	Road-650 Overstock	0.30	Excess Inventory	Reseller
10	ADVENTURE	Mountain Tire Sale	0.50	Excess Inventory	Customer

Conclusion:
